2) $V(h) = \frac{\pi}{3} \cdot h^2 \cdot (3r - h)$

$V = 477\ m^3$

$r = d/2 = 5\ m$

$475 = \frac{\pi}{3} \cdot h^2 \cdot (15 - h)$

$\pi \cdot h^2 \cdot 5 - \frac{\pi}{3} \cdot h^3 - 477$

$f(h) = \frac{\pi}{3} h^3 - 5\pi h^2 + 477$

$f'(h) = \pi h^2 - 10\pi h = \pi h (h - 10)$

$h_{n+1} = h_n - \frac{f(h_n)}{f'(h_n)}$

1: $9 - \frac{243\pi - 405\pi + 477}{81\pi - 90\pi} = 9 - \frac{(-162\pi) + 477}{-9\pi} = 9 - \frac{-37.9380}{-28.2743} = 7.6582\ m$

$|7.6582 - 7.3955|$ $0.2627 > 0.001$

2: $7.6582 - \frac{149.7127\pi - 293.2401\pi + 477}{58.6480\pi - 76.582\pi} = 7.6582 - \frac{-14.8213}{-56.3438} = 7.3955\ m$

$|7.3955 - 7.3808|$ $0.0747 > 0.001$

3: $7.3955 - \frac{-0.8874}{-60.5068} = 7.3808$

$|7.3808 - 7.3807|$ $0.0001 < 0.001$

4: $7.3808 - \frac{-0.0038}{-60.726} = 7.3807$

Wasser $h = 7.381\ m$